Embedded Air-Coupled Ultrasonic 3D Sonar System with GPU Acceleration

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***Abstract*—Air-coupled ultrasonic sonar systems are a valuable complement to lidar sensors for safe navigation of autonomous vehicles, especially in environments with optically transparent and reflective surfaces. Consequently, sonar systems based on phased arrays and beamforming are a promising compact solution, providing precise and unambiguous 3D localization of multiple objects in a wide field of view. However, a cost-effective and viable integration is challenging due to the increased component count and computational effort. Therefore, we present an embedded phased array sonar system requiring a minimum number of low-cost components, and providing high frame rates. The system contains 36 digital MEMS microphones, one 40 kHz transducer, an FPGA for CIC-filtering and communication with the main processor (Nvidia Jetson Nano) via Hi-Speed USB or WiFi. The GPU accelerated frequency-domain signal processing consisting of matched filtering, optimized conventional narrow-band beamforming and envelope extraction, enables frame rates up to 30 Hz. In an anechoic chamber, a field of view of $\pm 80^\circ$ and an angular resolution of 14° have been measured using two $\varnothing$100 mm spheres. Therefore, this self-contained phased array sonar allows a practicable sensor fusion with lidar data for a more reliable monitoring of the environment.**

I. Introduction

Autonomous ground and air vehicles rely on a combination of multiple sensor types with complementary strengths and weaknesses for robust monitoring of their environment [1]. For example, lidar sensors operate unreliably in harsh environments [2] and in the presence of optically reflective and transparent surfaces [3], as found typically in modern buildings. Therefore, as a complement, ultrasonic sensor systems are added [4]–[7], which are unaffected by these conditions.

In these studies, multiple 1D ultrasonic range finders are used, each covering a specific directional region, also referred to as sonar ring [8]–[10]. However, in order to achieve a wide field of view, this method requires many components and a lot of mounting space. In addition, the angular resolution is limited by the beam width of the transducer and there is no height differentiation.

A second type of sonar system relies on triangulation using one transmitter and three to five receivers [11]–[16]. Here, the direction of objects is calculated directly from the relative phase shifts of the echo signals, thus, requiring low computational effort and few components. However, the localization fails if multiple objects are at a similar range and in different directions, i.e. when the echo signals overlap.

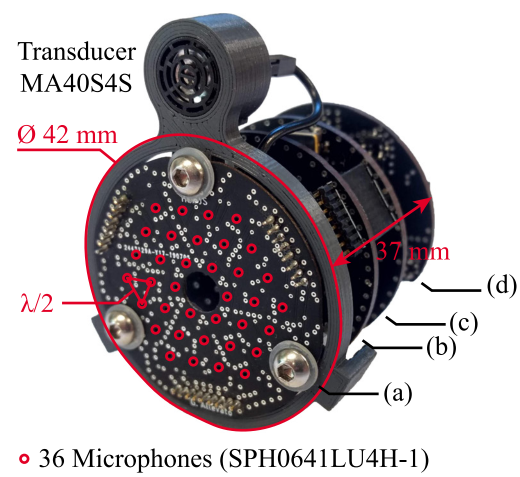

Fig. 1. The modular sonar head includes a stack of four PCBs containing 36 microphones (a), the driving electronics of the ultrasonic transducer (b), an FPGA for signal acquisition and conversion (c), and communication via WiFi or alternatively via Hi-Speed USB (d). This design simplifies the evaluation of new modules, but increases the overall size.

The third technique is based on beamforming using phased arrays of MEMS microphones and one or more transmitters [17]–[23], or, recently, fully digital MEMS transceivers [24]. This approach maintains a compact size and provides precise, unambiguous 3D localization of multiple objects with a wide field of view. Thus, it is a promising complement for improving not only obstacle avoidance, but also mapping and navigation. On the other hand, the increased component count and computational effort make a cost-effective and viable integration challenging.

Therefore, based on our previous MIMO phased array [25]–[29], we present a scaled-down embedded sonar system requiring a minimum number of components (Section II). Enabled by the advancements in powerful small-size computers, the optimized narrow-band signal processing is additionally GPU accelerated for fast image generation (Section III). An experimental characterization of the imaging capabilities and achieved frame rates are provided in Section IV.

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II. System Architecture

The system consists of the sonar head for pulse transmission and echo reception, and the Nvidia Jetson Nano as the main signal processor (Fig. 2). The sonar head includes a low-cost narrow-band ultrasonic transducer (Murata MA40S4S) with a diameter of 10 mm, driven with a bipolar square-wave signal (±10 V, 40 kHz). The driving signal is generated using a H-bridge MOSFET driver (Analog Devices ADP3630), controlled by an FPGA (Intel MAX10) and supplied by a 5 V-to-10 V step up converter (Analog Devices ADP1613).

For echo reception, we use an array of 36 MEMS microphones (Knowles SPH0641LU4H-1), arranged on an equilateral triangular $\lambda/2$ grid, forming a hexagonal aperture with a maximum dimension of 25.7 mm (Fig. 1). Compared to a 36-element uniform rectangular array (URA), this arrangement has reduced side lobes in the receive pattern of the conventional beamformer, whereas the main lobe width is increased by only 1°. The microphones are small, low-power, and broadband (10 Hz to 80 kHz). The integrated delta-sigma ADCs directly provide digital pulse density modulated (PDM) signals (1-bit, 4 MSa/s), thus additional components are not required. In addition, each pair of microphones shares its data line by launching the interleaved data on the rising and falling edge, reducing FPGA pin usage.

The FPGA contains 36 sinc-3 Cascaded-Integrator-Comb-filters (CIC) [30], converting and downsampling the 1-bit PDM signals to 16-bit pulse code modulated (PCM) signals sampled at a rate of 125 kSa/s.

There are two options for communication with the Jetson Nano. First, the FPGA is connected to a USB FIFO IC (FTDI FT232H) via an 8-bit parallel bus, providing a transfer rate of 480 Mbit/s for bidirectional communication. Since the array data rate is lower (72 Mbit/s) than the transfer rate, intermediate buffering is not required, further reducing the component count. The second option features a microcontroller (MCU) module (Espressif ESP32 Wrover-B) containing a WiFi-ready MCU and 8 MB of PSRAM. The FPGA is directly connected to the memory-mapped PSRAM, transferring data via QSPI with a transfer rate of 160 Mbit/s, whereas the UART interface is used to receive control commands. The bottleneck is the WiFi transfer rate of 14 Mbit/s on average using TCP/IP. Therefore, at the expense of frame rate, the sonar head is detachable from the main processor, useful for space constrained applications or for centralized processing of multiple sensors.

The Nvidia Jetson Nano is a small-size single board computer featuring a 1.4 GHz quad-core CPU and a Maxwell GPU with 128 CUDA cores, enabling parallel signal processing. Here, a cross-platform Qt application (C++) is executed for system control and signal processing for image generation.

III. Image generation

Image generation is composed of two processes, i.e. data acquisition and array signal processing. Both processes are run in parallel to increase the frame rate. First, in order to transmit an ultrasonic pulse, the transducer is excited with the driving signal (square-wave, 40 kHz, ±10 V) for 1 ms.

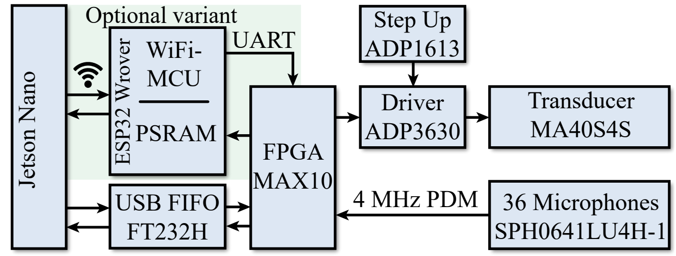

Fig. 2. Block diagram of the complete system. Since the microphone signals are digital and the transducer is driven with a simple square-wave signal (±10 V, 40 kHz), the sonar head requires a minimum number of components.

Second, scatterers in the frontal hemisphere of the sonar head reflect echos, recorded by the microphone array. Thus, the raw $[M \times N]$ signal matrix $\mathbf{S}$ is obtained and transferred to the Jetson Nano for signal processing, where M is the number of microphones and N is the sample count.

The array signal processing consists of frequency-domain analytic signal conversion, matched filtering, conventional beamforming [31], and envelope extraction using custom CUDA kernels and the parallel FFT library *cuFFT*. First of all, each row of $\mathbf{S}$ is transformed to the frequency domain. However, only the first half of each spectrum is calculated for analytic signal conversion. Then, each row of the transformed matrix is multiplied element-wise with the frequency response of the matched filter $\hat{\mathbf{h}}$, i.e.

$$\hat{\mathbf{S}}_{\mathrm{F},(m,*)} = \mathscr{F}_N \left\{ \mathbf{S}_{(m,*)} \right\} \odot \hat{\mathbf{h}} \quad \text{for } m = 0, \ldots, M-1. \quad (1)$$

Next, the resulting pre-processed matrix $\hat{\mathbf{S}}_{\mathrm{F}}$ is multiplied with the conventional $[K \times M]$ beamformer matrix $\mathbf{W}^{\mathsf{H}}$ to obtain the spatially filtered signal spectra for K different scanning directions. This operation is implemented as parallel tiled matrix multiplication to reduce processing time. For further optimization, only the frequency bins of $\hat{\mathbf{S}}_{\mathrm{F}}$ containing the narrow-band echo spectrum are considered for multiplication. Finally, each row of the spatially filtered matrix is transformed to the time-domain, and the amplitude envelope matrix $\mathbf{S}_{\mathrm{E}}$ is obtained by calculating the absolute value of each element, i.e.

$$\mathbf{S}_{\mathrm{E},(k,*)} = \left| \mathscr{F}_N^{-1} \left\{ \left[\mathbf{W}^{\mathsf{H}} \cdot \hat{\mathbf{S}}_{\mathrm{F}} \right]_{(k,*)} \right\} \right| \quad \text{for } k = 0, \ldots, K-1, \quad (2)$$

where a single element of $\mathbf{W}^{\mathsf{H}}$ is given by

$$\mathbf{W}^{\mathsf{H}}_{(k,m)} = \mathrm{e}^{-j2\pi \frac{f_0}{c} \left[x_m \sin(\theta_k) \cos(\varphi_k) + y_m \sin(\varphi_k) \right]}. \quad (3)$$

Here f_0 is the signal frequency, c is the speed of sound, (x_m, y_m) is the m-th element's position, and θ_k, φ_k are the azimuth and elevation angles of the k-th scanning direction.

Optional, the image is visualized in two or three dimensions using OpenGL by rendering the color-coded and alpha-blended amplitude values of $\mathbf{S}_{\mathrm{E}}$ at their corresponding position (Fig. 3).

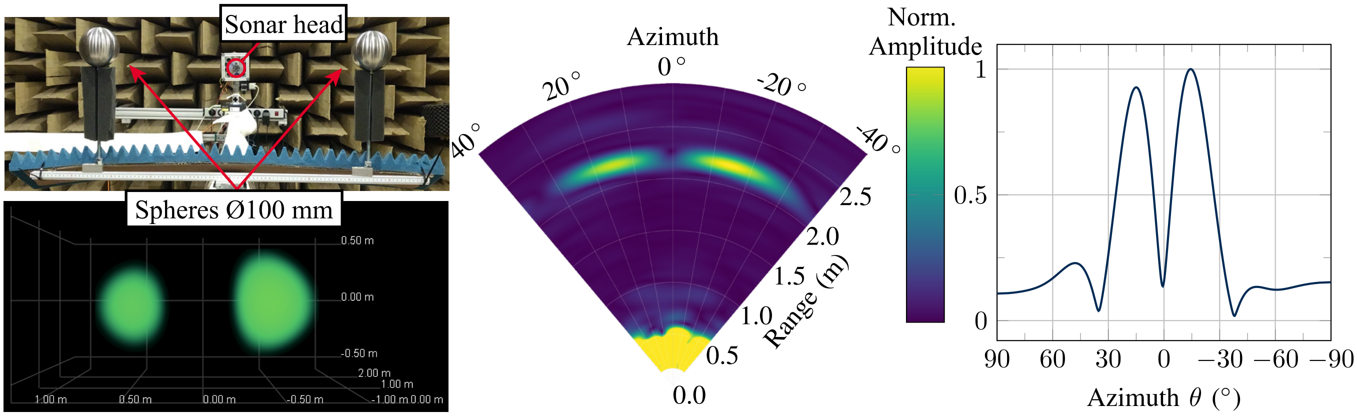

Fig. 3. Setup for measuring the angular resolution (top left). The distance between the spheres at a range of 2 m is gradually increased until they are separately detectable. Rendered 3D-Scan of the scene (bottom left). Corresponding horizontal sectional view (B-Scan) (middle) and the sectional view at 2.1 m (right). The echoes of the spheres create two distinct local maxima, thus they are separately detectable.

IV. Experimental Results and Discussion

We measured the system's range of view, field of view, angular resolution, and the achievable frame rates in an anechoic chamber. One or two hollow steel spheres (⌀100 mm) are used as targets, mounted on a movable slide. The sonar head is attached to two rotational axes at the end of the rail, so the targets can be positioned freely in the coordinate system (Fig. 3).

In order to measure the range of view, one sphere is positioned directly in front of the sonar head (0°, 0°), and the range is gradually increased. The ranges where the echo signal is below the noise level, i.e. not detectable, determine the minimum and maximum detection ranges, which are 0.17 m and 5.5 m, respectively. The field of view is determined using the same procedure, except the range is fixed at 2 m and the target's azimuth angle is gradually increased. The sphere is detectable in a field of view of ±80°, mainly depending on the directivity of the transducer.

The angular resolution is determined by the minimum angle between two objects in order to detect them separately. For separation, the corresponding echoes must create two distinct local maxima in the image. For measuring the angular resolution two spheres are positioned at a range of 2 m directly in front of the sonar head. The distance between the spheres is increased gradually. With a distance greater than 440 mm, the spheres are separately detectable, corresponding to an angular resolution of 14°. The measured value is in good agreement with the simulation (13°) using an analytical model.

The frame rate is determined by measuring and averaging (32×) the time between two completely generated frames. The total time is mainly composed of sound travel, data transfer and processing time. Using a fixed viewing range of 3 m (2048 samples), the frame rates over the number of scanning directions of the WiFi and USB variant are compared (Fig. 4). The USB variant is clearly superior due to the higher data transfer speed, thus allowing faster movements.

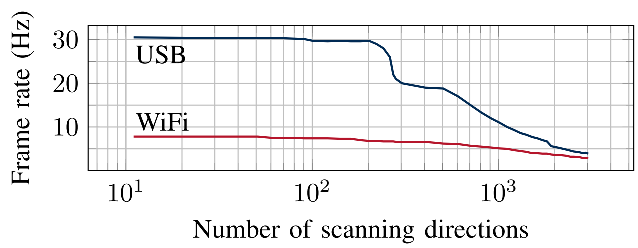

Fig. 4. The USB variant is limited by the data transfer rate up to 240 directions. The frame rate drops if the processing time is higher than the data transfer time, as the parallelization of the data acquisition and signal processing becomes out of sync. With increasing scan directions the achievable frame rate decreases for both variants. The WiFi variant is limited by the data transfer rate for all numbers of scan directions.

V. Conclusion and Future Work

We presented an embedded ultrasonic imaging system capable of localizing multiple objects in 3D with a wide field of view. The GPU acceleration and optimization of the frequency-domain signal processing ensure sufficient frame rates for monitoring dynamically changing environments, e.g. due to motion. Since the microphones provide digital data and the narrow-band transducer is driven with a single-frequency square-wave signal, the detachable sonar head requires only a minimum number of low-cost components. Therefore, its size can be further reduced to a single PCB for applications with tight space constraints. Although a drawback of the small aperture size and the beamforming technique is the limited angular resolution, it is sufficient in the primary operating range up to 3 m, e.g. to separately detect doorways for navigation. All in all, this self-contained imaging system can be viably integrated into e.g. mobile autonomous ground and air vehicles for improving obstacle avoidance, mapping and navigation. Future work will focus on the automated extraction of object coordinates, Doppler velocity estimation and the evaluation of the sensor fusion with lidar data.

REFERENCES

[1] M. B. Alatise and G. P. Hancke, "A Review on Challenges of Autonomous Mobile Robot and Sensor Fusion Methods," *IEEE Access*, vol. 8, pp. 39 830–39 846, 2020.

[2] A. Ziebinski, R. Cupek, H. Erdogan, and S. Waechter, "A Survey of ADAS Technologies for the Future Perspective of Sensor Fusion," in *Computational Collective Intelligence*. Cham: Springer International Publishing, 2016, vol. 9876, pp. 135–146.

[3] S.-W. Yang and C.-C. Wang, "On Solving Mirror Reflection in LIDAR Sensing," *IEEE/ASME Transactions on Mechatronics*, vol. 16, no. 2, pp. 255–265, Apr. 2011.

[4] H. Wei, X. Li, Y. Shi, B. You, and Y. Xu, "Fusing Sonars and LRF Data to Glass Detection for Robotics Navigation," in *2018 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. Kuala Lumpur, Malaysia: IEEE, Dec. 2018, pp. 826–831.

[5] R. Singh and K. S. Nagla, "Multi-data sensor fusion framework to detect transparent object for the efficient mobile robot mapping," *International Journal of Intelligent Unmanned Systems*, vol. 7, no. 1, pp. 2–18, Jan. 2019.

[6] N. Gageik, P. Benz, and S. Montenegro, "Obstacle Detection and Collision Avoidance for a UAV With Complementary Low-Cost Sensors," *IEEE Access*, vol. 3, p. 11, 2015.

[7] Xue-Cheng Lai, Cheong-Yeen Kong, Shuzhi Sam Ge, and A. Al Mamun, "Online map building for autonomous mobile robots by fusing laser and sonar data," in *IEEE International Conference Mechatronics and Automation, 2005*, vol. 2. Niagara Falls, Ont., Canada: IEEE, 2005, pp. 993–998.

[8] D. Browne and L. Kleeman, "An advanced sonar ring design with 48 channels of continuous echo processing using matched filters," in *2009 IEEE/RSJ International Conference on Intelligent Robots and Systems*. St. Louis, MO, USA: IEEE, Oct. 2009, pp. 4040–4046.

[9] D. Bank and T. Kampke, "High-Resolution Ultrasonic Environment Imaging," *IEEE Transactions on Robotics*, vol. 23, no. 2, pp. 370–381, Apr. 2007.

[10] K.-W. Jörg, "World modeling for an autonomous mobile robot using heterogenous sensor information," *Robotics and Autonomous Systems*, vol. 14, no. 2-3, pp. 159–170, May 1995.

[11] N. Seckel and A. Singh, "Physics of 3D Ultrasonic Sensors," Toposens GmbH, Tech. Rep., 2019, www.toposens.com.

[12] Kreczmer, "Estimation of the Azimuth Angle of the Arrival Direction for an Ultrasonic Signal by Using Indirect Determination of the Phase Shift," *Archives of Acoustics*, vol. 44, no. 3, pp. 585–601, 2019.

[13] F. Guarato, V. Laudan, and J. F. Windmill, "Ultrasonic sonar system for target localization with one emitter and four receivers: Ultrasonic 3D localization," in *2017 IEEE SENSORS*. Glasgow: IEEE, Oct. 2017, pp. 1–3.

[14] C. Walter and H. Schweinzer, "Locating of objects with discontinuities, boundaries and intersections using a compact ultrasonic 3D sensor," in *2014 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*. Busan, South Korea: IEEE, Oct. 2014, pp. 591–600.

[15] A. Diosi and L. Kleeman, "Advanced sonar and laser range finder fusion for simultaneous localization and mapping," in *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (IEEE Cat. No.04CH37566)*, vol. 2. Sendai, Japan: IEEE, 2004, pp. 1854–1859.

[16] B. Barshan and R. Kuc, "A bat-like sonar system for obstacle localization," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 22, no. 4, pp. 636–646, July-Aug./1992.

[17] R. Kerstens, D. Laurijssen, and J. Steckel, "eRTIS: A Fully Embedded Real Time 3D Imaging Sonar Sensor for Robotic Applications," in *2019 International Conference on Robotics and Automation (ICRA)*. Montreal, QC, Canada: IEEE, May 2019, pp. 1438–1443.

[18] J. Steckel, A. Boen, and H. Peremans, "Broadband 3-D Sonar System Using a Sparse Array for Indoor Navigation," *IEEE Transactions on Robotics*, vol. 29, no. 1, pp. 161–171, Feb. 2013.

[19] S. Kumar and H. Furuhashi, "Long-range measurement system using ultrasonic range sensor with high-power transmitter array in air," *Ultrasonics*, vol. 74, no. 74, pp. 186–195, 2017.

[20] K. Herman and L. Stach, "Analysis and Elaboration of an Air-Coupled Ultrasound Wideband Sensor Array," *Acta Physica Polonica A*, vol. 123, no. 1, p. 31, 2013.

[21] M. Turqueti, V. Kunin, B. Cardoso, J. Saniie, and E. Oruklu, "Acoustic sensor array for sonic imaging in air," in *2010 IEEE International Ultrasonics Symposium*. San Diego, CA, USA: IEEE, Oct. 2010, pp. 1833–1836.

[22] H. Furuhashi, Y. Uchida, and M. Shimizu, "Imaging sensor system using a composite ultrasonic array," in *2009 IEEE Sensors*. Christchurch, New Zealand: IEEE, Oct. 2009, pp. 1467–1472.

[23] M. Clapp and R. Etienne-Cummings, "Ultrasonic bearing estimation using a MEMS microphone array and spatiotemporal filters," in *2002 IEEE International Symposium on Circuits and Systems. Proceedings (Cat. No.02CH37353)*, vol. 1. Phoenix-Scottsdale, AZ, USA: IEEE, 2002, pp. I–661–I–664.

[24] S. Anzinger, F. Lickert, A. Fusco, G. Bosetti, D. Tumpold, C. Bretthauer, and A. Dehe, "Low Power Capacitive Ultrasonic Transceiver Array for Airborne Object Detection," in *2020 IEEE 33rd International Conference on Micro Electro Mechanical Systems (MEMS)*. Vancouver, BC, Canada: IEEE, Jan. 2020, pp. 853–856.

[25] G. Allevato, J. Hinrichs, M. Rutsch, J. Adler, A. Jäger, M. Pesavento, and M. Kupnik, "Real-Time 3D Imaging using an Air-Coupled Ultrasonic Phased-Array," *IEEE Transactions on Ultrasonics, Ferroelectrics and Frequency Control*, p. 11, 2020, early access June.

[26] G. Allevato, J. Hinrichs, D. Grosskurth, M. Rutsch, J. Adler, A. Jäger, M. Pesavento, and M. Kupnik, "3D imaging method for an air-coupled 40 kHz ultrasound phased-array," in *International Congress on Acoustics*, Aachen, 2019, p. 8.

[27] A. Jäger, D. Großkurth, M. Rutsch, A. Unger, R. Golinske, H. Wang, S. Dixon, K. Hofmann, and M. Kupnik, "Air-coupled 40 kHz Ultrasonic 2D-Phased Array Based on a 3D-printed waveguide structure," in *IEEE International Ultrasonics Symposium*, 2017, p. 4.

[28] C. Haugwitz, A. Jäger, G. Allevato, J. Hinrichs, A. Unger, S. Saul, J. Brotz, B. Matyschok, P. Pelz, and M. Kupnik, "Flow Metering of Gases Using Ultrasonic Phased-Arrays at High Velocities," in *IEEE International Ultrasonics Symposium (IUS)*, Glasgow, 2019, p. 4.

[29] A. Jäger, J. Hinrichs, G. Allevato, M. Sachsenweger, S. Kadel, D. Stasevich, W. Gebhard, G. Hubschen, T. Hahn-Jose, W. M. D. Wright, and M. Kupnik, "Non-contact ultrasound with optimum electronic steering angle to excite Lamb waves in thin metal sheets for mechanical stress measurements," in *IEEE International Ultrasonics Symposium (IUS)*, Glasgow, 2019, p. 4.

[30] E. Hogenauer, "An economical class of digital filters for decimation and interpolation," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 29, no. 2, pp. 155–162, Apr. 1981.

[31] A. M. Zoubir, M. Viberg, R. Chellappa, and S. Theodoridis, Eds., *Array and Statistical Signal Processing*, first edition ed., ser. Academic Press Library in Signal Processing. Amsterdam: Academic Press, 2014, no. 3.